

Planning Worksheets

Edit this file and save it as a PDF and submit a single .pdf file to the assignment drop box. Both you and your partner must submit a copy of this completed PDF to Brightspace. The file you submit should be named `comp1405_f23_###_###_assignment_08.pdf` with `###` replaced by the last three digits of each partner's Student ID number. If you are working alone, only put your own Student ID number.

Brightspace. Step 0: Features

What features have you and your partner selected? Remember to choose one from each category.

1. **Category 1:** Add rooms to a file
2. **Category 2:** Colour Anomaly

Step 1: User Stories

For both of your selected features, write **one to two User Stories each** which express the main value of the features. Follow the format seen in class: "As a **<role>**, I want to **<action>**, so that I can **<motivation>**"

1. As a user, I want to input a text file with rooms of my choice, so that I can use those rooms in the game.
2. As a user, I want to change given colours with colours of my choice, so that I can use this new colour anomaly in the game.

Step 2: Requirements & Constraints

Carefully consider your feature selections. Review the code, the provided documentation, and the feature description to write a list of required functionality and constraints that must be considered. Consider cases where the feature might not work as intended, changes to the data to show your feature, or techniques you cannot use. As an example, one constraint for all features is that **the Duty.py module cannot be modified**.

Category 1

1. **Constraint:** Cannot modify the Duty.py module
2. The user must type the room details in the proper format: Master Bedroom-Bed,Sofa,Television
- 3.

Category 2

1. **Constraint:** Cannot modify the Duty.py module
2. If the user has inserted room with an object that is a colour that is not present in the original list, it will not change
3. Must make sure that the order of the list does not change, otherwise it will give number anomaly

Step 3: Prioritize

While you may not have many User Stories, with your partner, determine the best order to complete these features and write the order below. The highest priority will be at the top in position 1.

Prioritized User Stories

1. For Add Room: decide format of input file, access input file with RunGame.py, decide how you are going to separate the room name and items in the room to use the addRoom() function
2. For Colour Anomaly: create a list with current colours in the program, create a list with new colours of your choice, loop through the items in the room, if a colour is present, replace it with any of the new colours.

For each User Story, justify why you and your partner prioritized it here. **Be brief – one to two sentences maximum.**

1. We must make sure the user inputs a correct formatted text file as the program is designed to expect this particular format otherwise the room will not be added.
2. We will need a list of the current colours as the program is going to be searching for these colours when we want to change them into the new colours we have selected.

Step 4: Task List

For each user story, write a **task list**. In real work, tasks could be as large as the user stories, but in this class we will work at a smaller scale. For each user story, write a **list of tasks** which cover everything that your team will need to do to write the code, make sure it works, and submit in Assignment 9. Review the code in **RunGame.py** to see what steps need to be accomplished to do similar work to your features. Assume that you will be starting with the **RunGame.py** code and do not need to include tasks that are already included there. Include any functions that you will need to review or likely need to use in the task description. This is **not** an algorithm; it is a checklist for you and your partner.

Category 1 Task List(s)

1. Select the format of the text file, particularly, how the user should write the new room
2. Make sure the file is accepted as a command line argument
3. Access the file in code to see the contents of it
4. Find a way to separate the room name and the room items in order to use the addRoom() function in Duty

Category 2 Task List(s)

1. Create a list of the current colours in RunGame.py
2. Create a list of the new colours you like in RunGame.py
3. Find a way to access the list of items in the room
4. Find a way to loop through the list of items and check if any of the current colours are present
5. Change the current colour to any of the new colours
6. Make sure the items are still in correct order and return the new list by using shiftingElements(theList, firstIndex, secondIndex) function
7. Also use Duty.addAnomaly(self, name:str, room:Room, modified_item_list:list[str])->bool

Team Worksheet

Finally, this section is here to keep your team on track. It will receive full marks if you are registered in a group on Brightspace, and a reasonable effort was made to respond to each point. If you are working individually, you will not receive marks for this section.

Team Member Names:

1. Maansinh Jadeja
2. Abdulrahman Abdelrazeq

What days are you planning to meet to work together to code Assignment 9, in-person or online?

Online on Monday, Wednesday and Thursday.

How are you planning to pair-program together (e.g., screen sharing, together in-person, VS Code Live Share)?

Discord share screen

How long do you each anticipate working on the code for this assignment?

Maansinh – As long as it needs to take in order to finish a minimum of 1 task per day (1-2 hours per day)
Abdulrahman – As long as it needs to finish a minimum of 2 tasks per day (1-2 hours per day)

Who will be submitting this PDF and who will be submitting the Assignment 9 Python file?

Both group members will have to submit the PDF file and the assignment 9 python file along with the text file for the category 1 feature.

Remember to submit your responses as a PDF on Brightspace and ensure that **both group members have submitted**. You **must** have registered your group before the group registration deadline in order to receive marks for this assignment if you are working in a group.

BONUS: Function Planning

This section is optional, but can provide **10% bonus marks**, up to a **maximum of 100%** on this assignment to make up for other lost marks, if completed. Bonus marks will be provided if an attempt is made to answer these questions on either a per-task or per-feature basis, whether working individually or as a team. Total coverage of all questions across all tasks is not strictly required, but aim to complete this as best as you can to better coordinate with your partner and have a clear picture of the work required for Assignment 9.

For each task in your task lists, try to write out and plan the following:

1. What functions from the **Duty.py** module will you need to use?

Category 1: addRoom(self, name: str, items: list[str])→bool

Category 2: addAnomaly(self, name:str, room:Room, modified_item_list:list[str])→bool

2. What functions from the **RunGame.py** file can you, and must you, modify or use as an example?

Category 1: We do need to use any function from RunGame.py

Category 2: We will modify, register_anomalies() and create_anomaly() to add the new colour anomaly with register_anomalies() and call it with create_anomaly() in the game

3. What functions should you write to cleanly separate our code into small pieces?

- a. What could you name the function?

- b. What is that function's one purpose?

- c. What parameters will that function accept, and what data type are they?

- d. Will the function return anything, and if so, what data type will it be and how will that return value be used?

- e. What would an example of using this function look like?

Category 1:

- (a) addExtraRoomWithInputFile()

- (b) Access the input file as the command line argument and add the rooms present in the file into the game.

- (c) No parameters accepted

- (d) No return values

- (e) User inputs the file and calls this function in order for them to add their rooms into the game.

Category 2:

- (a) addNewColourAnomaly()

- (b) To create the colour anomaly and add it into the game

- (c) No parameters accepted

- (d) No return values

- (e) It will access all the rooms and check for existing colours and replace them with new colours

Some tasks may not be appropriate for these questions. Try to come up with similar questions for your tasks.

For each feature, design a small algorithm by writing either a flowchart or a simple text algorithm which expresses the flow of events without worrying about the Python syntax.

Write your responses either here or on later pages.